

CS & IT ENGINEERING



Computer Network

Network Address Translation

Lecture No. - 08



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Recap of Previous Lecture



Topic

CIDR Allocation





Topics to be Covered



Topic

Network Address Translation





Topic : IPv4 Address

$\Rightarrow$ must be unique



→ IPv4 address (32-bits)

↓
Range 0 to $(2^{32} - 1)$
Unsigned
(Max^m) 2^{32} IP Address



Topic : IPv4 Address



→ IPv4 address (32-bits) have range problem.



Topic : IPv4 Address



→ Solution for IPv4 address (32-bits) range problem.

1. Network Address Translation (NAT)

[Short-term solution]

2. IPv6 address (128 bits)

[Permanent solution]

→ (2^{128} IP Address)



Topic : NAT

→ NAT : Network Address Translation

→ Internet : Public Network →

→ Every network is considered as a private network

* Internet
→ largest WAN

* Router
→ WAN device





Topic : NAT



Private network



→ Every connected network is identified by unique public IPv4 Address
[Assigned by ISP]



→ Total number of network can be exist (world wide) is 2^{32}



Topic : NAT



Private
Network

→ All hosts inside a network is identified by private IPv4 Address



Topic : Private IPv4 Address

→ Network addresses for private IPv4 Networks :

10.0.0.0 / 8

172.16.0.0 / 12

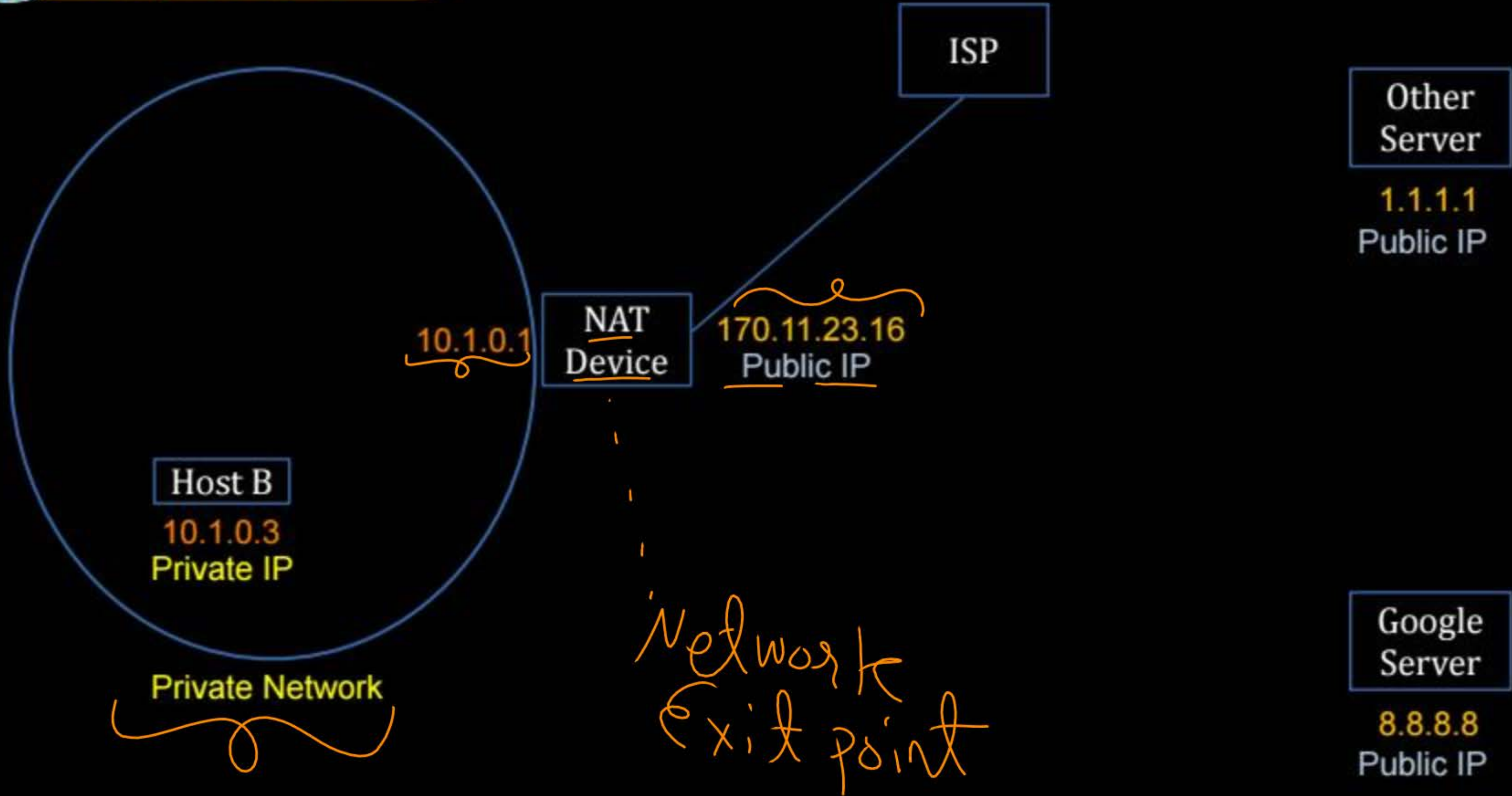
192.168.0.0 / 16

→ Mostly college/organization
* with 8 bit subnetting (16)
Mask → 16 one's = 255.255.0.0

→ Mostly in home (small) network
With 8 bit subnetting (24)
Mask → 24 one's = 255.255.255.0



Topic : NAT





Topic : NAT



→ Source host → Dest. host
→ Host B is sending one IP packet to Google server (NAT Device)

Source IP Address : 10.1.0.3
[Private IP]

170.11.23.16
(Public IP) ✓

Destination IP Address : 8.8.8.8



Topic : NAT Table



→ NAT device maintain,

NAT table for address translation of incoming datagram



Topic : NAT Table



<u>Local Private</u> <u>IPv4 Address</u> [Source IP Add.]	<u>Global Public</u> <u>IPv4 Address</u> [Destination IP Add.]
→ <u>10</u> . <u>1</u> . <u>0</u> . <u>3</u>	<u>8</u> . <u>8</u> . <u>8</u> . <u>8</u> →



Topic : NAT



→ Source host → Dest. host
→ Google server is sending one IP packet to Host B

Source IP Address : 8 . 8 . 8 . 8 ✓

Destination IP Address : 170 . 11 . 23 . 16
[Public IP]

10.1.0.3
(private IP)

NAT Device



Topic : NAT Device



Private N/w to Public N/w

→ For every outgoing datagram,

it modify Source IP address from private IP address to public IP address





Topic : NAT Device



Public N/w to Private N/w

→ For every incoming datagram,

it modify Destination IP address from public IP address to private IP address



[MSQ] [GATE-2024, Set-1] [1-Mark] 115C



#Q. Which of the following fields is/are modified in the IP header of a packet going out of a network address translation (NAT) device from an internal network to an external network?

☒ A Header Checksum

☒ B Source IP → Private IP to Public IP

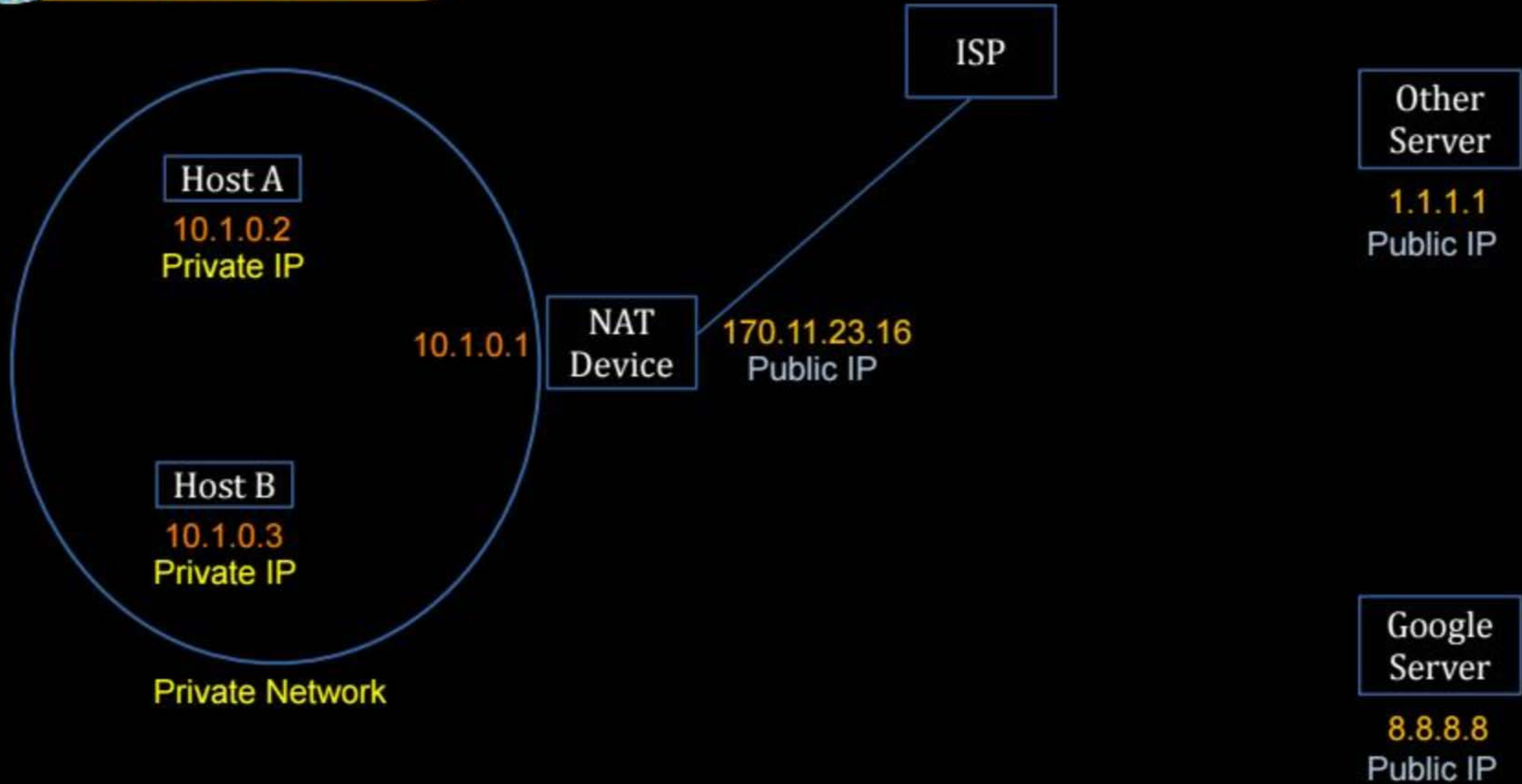
☐ C Destination IP

☐ D Total Length

Ans: A & B



Topic : NAT





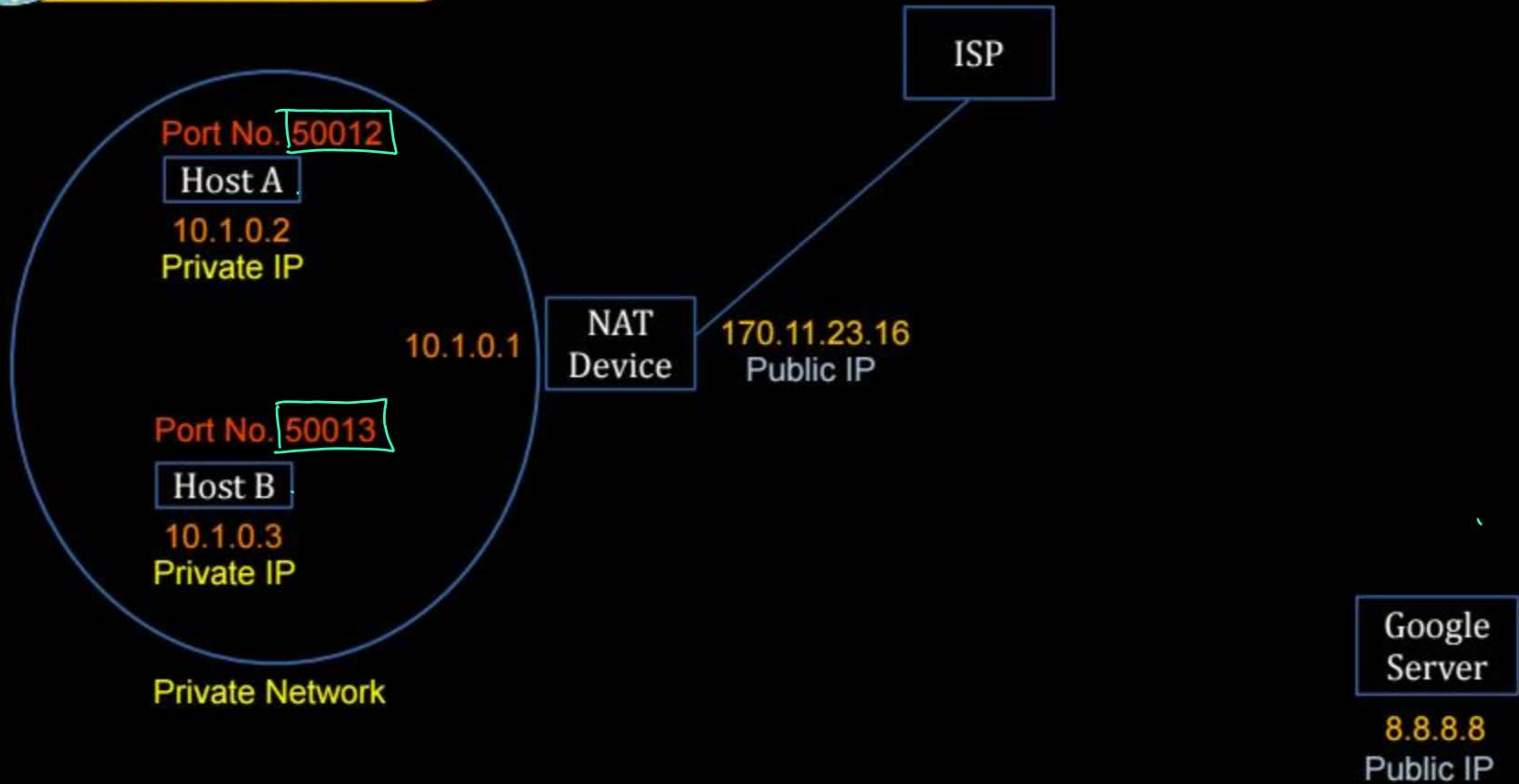
Topic : NAT Table



Local Private IPv4 Address [Source IP Add.]	Global Public IPv4 Address [Destination IP]
10 . 1 . 0 . 2	1 . 1 . 1 . 1
10 . 1 . 0 . 3	8 . 8 . 8 . 8



Topic : NAT





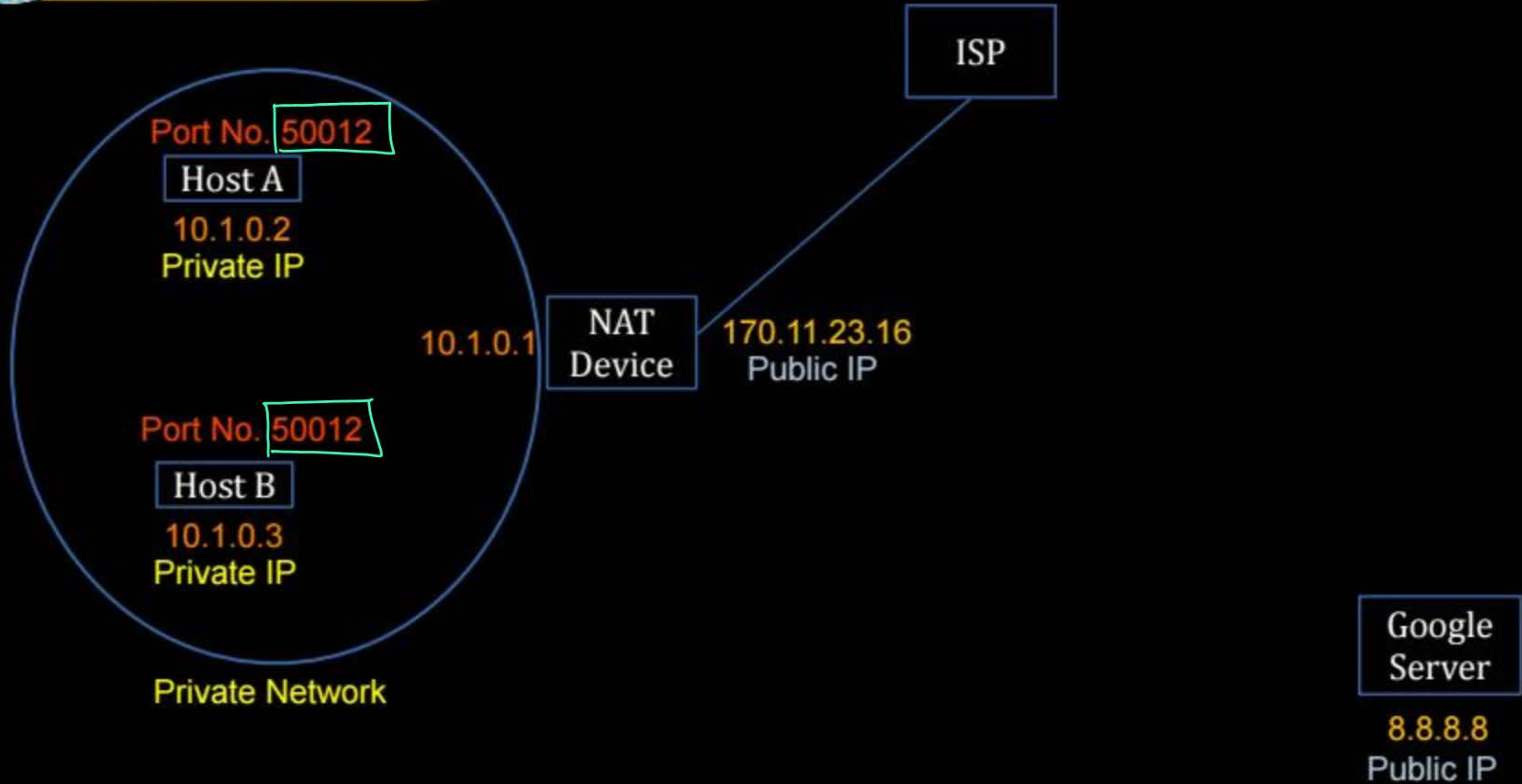
Topic : NAT Table



Local Private IPv4 Address [Source IP Add.]	Global Public IPv4 Address [Destination IP]	Source Port Number
→ 10.1.0.2	→ 8.8.8.8	→ 50012
→ 10.1.0.3	→ 8.8.8.8	→ 50013



Topic : NAT





Topic : NAT Table



⇒ PAT
Port Address
Translation



Local Private IPv4 Address [Source IP Add.]	Global Public IPv4 Address [Destination IP]	Source Port Number	Modified Source Port Number
→ 10.1.0.2 →	8.8.8.8	500 <u>12</u>	500 <u>12</u>
→ 10.1.0.3 →	8.8.8.8	500 <u>12</u>	500 <u>20</u>



Topic : NAT Device



→ As per requirement, NAT device can modify
Source Port Number field for outgoing packet
and Destination Port Number field for incoming packet



Topic : NAT Device



→ NAT device update address field of every outgoing and incoming IP packet



Topic : NAT Device



→ Mapping **internal IP addresses and ports** to **external IP addresses and ports**



Types of NAT :-



① static NAT → for server side
(fixed mapping)

② Dynamic NAT → for client side
* Pool of public IPs
(Dynamic Mapping)

③ PAT



2 mins Summary



Topic

Network Address Translation



THANK - YOU